

Question		Solution	Marks
4	(a)	Consider the body moving with initial velocity u with a constant acceleration a through a displacement s on a horizontal frictionless surface due to F. From the kinematics equation $v^2 = u^2 + 2as$ $s = \frac{v^2 - u^2}{2a}$ Work done by $F = Fs$ Since Gain in kinetic energy = Work done by F $\Delta E_K = Fs = mas$ $\Delta E_K = ma \frac{v^2 - u^2}{2a}$ $\Delta E_K = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$ Given that the object started from rest, its change in kinetic energy is its final kinetic energy OR its initial kinetic energy is zero, $E_K = \frac{1}{2}mv^2$	B1 B1 B1
	(b)	(i) $\Delta E_P = mgR \cos(145^\circ - 90^\circ)$ Loss in E_P = work done against friction (spelled in full) $mgR(\cos 55^\circ) = F \left(\frac{145}{180} \times \pi R \right)$ $F = 6.67 \text{ N}$	C1 B1 A1
		(ii) Work done against friction from P to the lowest point of the track $= F \left(\frac{\pi R}{2} \right)$ Increase in E_K = Decrease in E_P – Work done against friction $E_K = mgR - F \left(\frac{\pi R}{2} \right)$ $E_K = (3.0)(9.81)(2.0) - (6.67) \left(\frac{\pi \times 2.0}{2} \right)$ $E_K = 37.9 \text{ J}$	C1 A1

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5	(a)		$W = P \Delta V$	